PURE MATHEMATICS (P425/1)

1. Solve the equation: $\sqrt{4x+13} - \sqrt{x+1} = \sqrt{12+x}$

2. Solve the simultaneous equations below using the echelon form:

$$2x + 3y - 4z = 9$$
$$3x + 2y + z = 4$$
$$5x + y + 3z = 3$$

3. Solve for x : $9^{3x} - 6(9^{2x}) + 11(9^x) - 6 = 0$

4. Solve for x given: $0.7 = (1.2)^{2x-1}$

5. Solve the equations: $\log_2 x^2 + \log_2 y^3 = 1$

$$\log_2 x - \log_2 y^2 = 4$$

6. The roots of the quadratic equation $x^2 + (7 + p)x + p = 0$ are α and β .Given that α and β differ by 5, find the possible values of p

7. When the quadratic expression $px^2 + qx + r$ is divided by $x - 1$,$x - 2$ and $x + 1$, the remainders are 1,1 and 25 respectively. Determine the values of p, q and r *and* then factorise the expression completely.

8. When a polynomial $P(x)$ is divided by $x - 2$,the remainder is 4 and when $P(x)$ is divided by $x - 3$,the remainder is 7.Find the remainder when $P(x)$ is divided by $(x - 2)(x - 3)$

9. Using the substitution $u = 3^x$, solve the equation $3^x + 3^{2x} = 3^{3x}$; giving your answer correct to 3S.Fs

10. Find the value of x in the following equations;

    (a) $log_{10}(x + 9) = 2 + log_{10}x$

    (b) $1/9^x = 3^{x-2}$

11. Prove that : $\tan(45^0 + \vartheta) - \tan(45^0 - \vartheta) = 2\tan 2\vartheta$

12. Solve for x given $7\cos x + 24\sin x = 10$ and $0 \le x \le 360^0$.

13. a) Solve the simultaneous equations : $\cos x + 4\sin y = 1$ 4
$\sec x - 3\text{cosec}\, y = 5$ for $0^0 \le x,y \le 360^0$

    (b) Determine the solution of the equation $\tan 2x + 2\sin x = 0$ for $0 \le x \le 90^0$.

14. Prove the following identities

    (a) $2\text{cosec}2\vartheta = \text{cosec}\vartheta \sec\vartheta$

(b) $\frac{1+\tan^2 \theta}{2-\tan^2 \theta} = \sec 2\theta$

c) $\tan 5\vartheta = \frac{5\tan\vartheta - \tan^3\vartheta + \tan^5\vartheta}{1 - 10\tan^2\vartheta + \tan^4\vartheta}$

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